

The reduction potential is a measure of an oxidized molecule's affinity for electrons in a redox half-reaction. The table shows the standard reduction potential of several cytochromes involved in electron transport in the mitochondrial inner membrane.

Redox half-reaction	E° (V)
Cytochrome <i>a</i> (Fe ³⁺) + e ⁻ → Cytochrome <i>a</i> (Fe ²⁺)	0.29
Cytochrome <i>a</i> ₃ (Fe ³⁺) + e ⁻ → Cytochrome <i>a</i> ₃ (Fe ²⁺)	0.35
Cytochrome <i>b</i> (Fe ³⁺) + e ⁻ → Cytochrome <i>b</i> (Fe ²⁺)	0.08
Cytochrome <i>c</i> (Fe ³⁺) + e ⁻ → Cytochrome <i>c</i> (Fe ²⁺)	0.25
Cytochrome <i>c</i> ₁ (Fe ³⁺) + e ⁻ → Cytochrome <i>c</i> ₁ (Fe ²⁺)	0.22

According to the information, which series shows the sequence in which electrons pass through the cytochromes during oxidative phosphorylation?

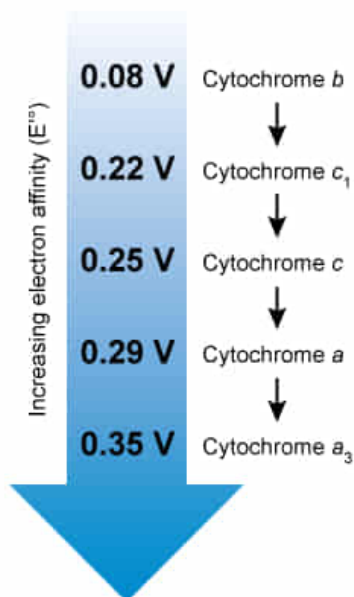
- ☐ A. Cytochrome *a* → cytochrome *a*₃ → cytochrome *b* → cytochrome *c*₁ → cytochrome *c* [2%]
- ☐ B. Cytochrome *a*₃ → cytochrome *a* → cytochrome *c* → cytochrome *c*₁ → cytochrome *b* [38%]
- ☐ C. Cytochrome *a*₃ → cytochrome *a* → cytochrome *b* → cytochrome *c*₁ → cytochrome *c* [4%]
- ✔ ☒ D. Cytochrome *b* → cytochrome *c*₁ → cytochrome *c* → cytochrome *a* → cytochrome *a*₃ [55%]

Correct

55% answered correctly

Explanation:

Electrons flow from cytochromes with low reduction potential to those with high reduction potential



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The **electron transport chain** (ETC) in the inner mitochondrial membrane generates the **proton gradient** required for **ATP synthesis** to occur. The electron transfer reactions of the ETC are carried out in an orderly sequence determined by the relative affinities of the ETC components for electrons. Components with less affinity donate electrons to components with greater affinity.

The reduction potential (E°) expresses the affinity of an oxidized molecule for electrons in a redox half-reaction under standard conditions. As reduction potential increases, the affinity of a molecule for electrons increases, and therefore electrons flow from

molecules with low E'° to molecules with high E'° .

The table in the question shows E'° for electron transfer half-reactions involving cytochromes in the mitochondrial ETC, and asks for the sequence in which electrons transfer between the cytochromes during oxidative phosphorylation. Because reduction potential indicates an oxidized molecule's affinity for electrons, **electrons are transferred** from species **with lower reduction potentials** (ie, less affinity for electrons) **to** species with **higher reduction potentials** (ie, more affinity for electrons) **throughout the chain**. Therefore, the correct sequence is cytochrome $b \rightarrow$ cytochrome $c_1 \rightarrow$ cytochrome $c \rightarrow$ cytochrome $a \rightarrow$ cytochrome a_3

(Choices A and C) The cytochromes involved in the ETC were not named for the order in which they handle electrons. The sequences in these available choices would require electrons to be transferred from molecules with greater electron affinity to molecules with less electron affinity (eg, from cytochrome a_3 to cytochrome b , or from cytochrome a to cytochrome b), which does not occur spontaneously.

(Choice B) Electrons are transferred from molecules with less affinity (ie, lower E'°) to molecules with greater affinity (ie, higher E'°). Therefore, this sequence is the *reverse* of the actual sequence of spontaneous transfers that occur between cytochromes in the ETC.

Educational objective:

The sequence of electron transfers that occurs in the electron transport chain is determined by the relative electron affinities of the molecules involved. The reduction potential is a measure of an oxidized molecule's affinity for electrons in a redox half-reaction.

Time spent: 0 sec

Subject:
Biochemistry

QID: 403335

System:
Metabolic Reactions